

Intelligent Document Processing on Azure

A Consultative Guide to Document Intelligence v4.0 & Competitive Strategy



The Dual Strategy: Balancing Precision with Reasoning



Pillar 1: Azure Document Intelligence (Traditional ML)

- **Best for:** Structured/Semi-structured forms, consistent layouts, high-speed batch processing.
- **Tech:** Neural networks & OCR.
- **Cost:** Lower cost/page.



Pillar 2: Azure Content Understanding (Generative AI)

- **Best for:** Unstructured docs, variable layouts, multimodal inputs (video/audio/images).
- **Tech:** GenAI with reasoning capabilities.
- **Key Advantage:** 'Zero-shot learning'—eliminates the need for training data on variable layouts.

Critical Insight:

Content Understanding uses generative AI to eliminate the dependency on document structure. It can extract professional experience from 10 different CV layouts with zero training, where traditional ML would struggle.

Benchmarking the IDP Landscape

Vendor	Accuracy (Structured)	Setup Effort	Pricing Model	Core Strength
Microsoft (Azure)	94% / High (GenAI ✨)	Zero Labeling (GenAI) or Labeled	Pay-per-page (Flexible)	<u>Hybrid Approach (ML + GenAI)</u>
AWS Textract	92%	Minimal	Pay-per-page + Free Tier	Developer-centric, Lambda Integration
Google Doc AI	94% (Best OCR)	Minimal	Pay-per-page + Credit	Handling Degraded Docs
ABBYY Vantage	90-98%	Pre-trained Skills	Annual Subscription (3-yr commit)	Legacy Enterprise Compliance

Key Takeaway: Azure and Google lead in accuracy (94%), while AWS offers deep infrastructure integration but trails slightly in structured accuracy.

Strategic Triggers for Vendor Selection

Choose AWS Textract If...

- Heavy investment in AWS infra (Lambda/Step Functions).
- Requirement for deep developer programmatic control.
- Primary need is high-volume asynchronous batch processing.



Choose Google Document AI If...

- Native GCP operation.
- Cost-efficiency for basic OCR is paramount.
- Need best-in-class handling of degraded documents (watermarks/noise).



Choose ABBYY Vantage If...

- Legacy on-prem needs.
- Massive volume (millions of pages) where subscription beats pay-per-page.
- Low-code platform needed for business users.



Choose Microsoft Azure If...

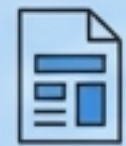
- Hybrid strategy needed (GenAI for reasoning + ML for speed).
- Requirement for extensive prebuilt models (Tax, Mortgage, ID).
- Leverage Power Automate/Microsoft 365 ecosystem.



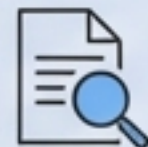
Azure Document Intelligence v4.0: The Core Toolkit



Semantic Extraction
Prebuilt & Custom Models.



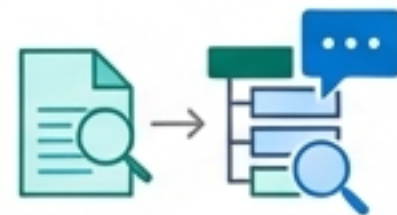
Layout Analysis
Structure, Tables, Selection Marks.
New: Office files, Markdown output.



Read (OCR)
Print & Handwriting. New: Searchable
PDF, Russian/Arabic/Thai support.

Feature Spotlight:

Retrieval-Augmented
Generation (RAG) readiness via
Markdown output and Section
Heading detection.



NEWS TODAY
Latest news and bulletin updates

Role
title
NEWS TODAY Latest news and bulletin updates

Content
139, 9, 608, 8, 608, 89, 199, 90

Polygon
139, 9, 608, 8, 608, 89, 199, 90

analyzeResult
{
 "apiVersion": "2022-08-31",
 "modelId": "gpt4o-mini-2024-07-18",
 "stringIndexType": "utf16CodeUnit",
 "content": "Tuesday, Sep 26, 2023 11:39:33 AM NEWS TODAY Latest news and bulletin updates",
 "pages": [
 {
 "pageNumber": 1,
 "angle": 0,
 "width": 736,
 "height": 1006,
 "unit": "pixel",
 "words": [
 {
 "text": "succeeded",
 "x": 139, "y": 9, "width": 608, "height": 89,
 "angle": 0,
 "unit": "pixel",
 "confidence": 1.0
 },
 {
 "text": "ste7108",
 "x": 139, "y": 9, "width": 608, "height": 89,
 "angle": 0,
 "unit": "pixel",
 "confidence": 1.0
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 "x": 139, "y": 9, "width": 608, "height": 89,
 "angle": 0,
 "unit": "pixel",
 "confidence": 1.0
 }
]
 }
]
}

Accelerating Financial & General Operations

CONTOSO LTD. INVOICE

Contoso (United States)
10000 1st St.
New York, NY 10001

Microsoft Corp
One Microsoft Way
Redmond, WA 98072

Invoice Number: 10000000000000000000
Invoice Date: 10/26/2023
Invoice Period: 10/01/2023 - 10/26/2023

DESCRIPTION	QUANTITY	UNIT PRICE	TAXES	TOTAL
Microsoft 365 E3	10	\$100.00	\$10.00	\$110.00
Microsoft 365 E3	10	\$100.00	\$10.00	\$110.00
Microsoft 365 E3	10	\$100.00	\$10.00	\$110.00

THANK YOU FOR YOUR BUSINESS!

CONTOSO BANK

Account Number: 12345678901234567890
Account Type: Checking
Statement Period: 10/01/2023 - 10/26/2023

Beginning Balance: \$1,000.00
Ending Balance: \$1,000.00

DATE	DESCRIPTION	AMOUNT
10/01/2023	Beginning Balance	\$1,000.00
10/05/2023	Deposit	\$100.00
10/10/2023	Withdrawal	\$50.00
10/15/2023	Withdrawal	\$50.00
10/20/2023	Deposit	\$100.00
10/25/2023	Withdrawal	\$50.00
10/26/2023	Ending Balance	\$1,000.00

Contoso

Item: 10000000000000000000
Item Type: 10000000000000000000
Item Price: \$100.00
Item Tax: \$10.00
Item Total: \$110.00

DATE: 10/26/2023
TIME: 05:00 PM
DEPT: 10000000000000000000
TIME: 02:00 PM
Folio Number: 1234567890

DATE	DESCRIPTION	CHARGES	CREDITS
10/26/2023	Room 1000	\$100.00	
10/26/2023	Tax 1000	\$10.00	
10/26/2023	Room 1000		\$100.00
10/26/2023	Tax 1000		\$10.00
10/26/2023	Room 1000		\$100.00
10/26/2023	Tax 1000		\$10.00
10/26/2023	Room 1000		\$100.00
10/26/2023	Tax 1000		\$10.00

Total: \$110.00
Balance: \$110.00

Contoso Bank

Check Number: 10000000000000000000
Check Date: 10/26/2023
Check Amount: \$100.00

Pay To the Order of: Contoso Ltd.
Amount in Words: One Hundred and 00/100 DOLLARS

Signature: [Signature]

General: Invoice

Extracts customer, vendor, billing address, amount due, tax details.

Financial: Bank Statement

Extracts beginning/ending balances and full transaction lists for reconciliation.

General: Receipt

Extracts merchant, transaction date, total, tax, tip. New: Hotel receipt support.

Financial: Checks

Extracts payer, payee, amount (written/numerical), date, signature.

Specialized Vertical: US Mortgage Processing

The image displays a collage of five standardized mortgage forms used in the US processing pipeline:

- 1003 Uniform Residential Loan Application:** A form for borrowers to provide personal and financial information, including income, assets, and debts.
- 1004 Uniform Residential Appraisal Report:** A form for appraisers to provide a professional opinion of the property's value, including a description of the property and a list of comparable sales.
- 1005 Request for Verification of Employment:** A form for lenders to request verification of a borrower's employment history and salary from the employer.
- Closing Disclosure (Final terms/costs):** A form that provides the borrower with the final terms of the loan, including the interest rate, monthly payment, and closing costs.
- Closing Disclosure (Loan terms, projected payments):** A form that provides the borrower with the final terms of the loan, including the interest rate, monthly payment, and projected payments.

The "100-Series" Suite:

- 1003 Uniform Residential Loan Application
- 1004 Uniform Residential Appraisal Report
- 1005 Verification of Employment
- 1008 Underwriting Summary
- Closing Disclosure (Final terms/costs)

Supports Fannie Mae and Freddie Mac requirements.

Closing Disclosure: Loan terms, projected payments.

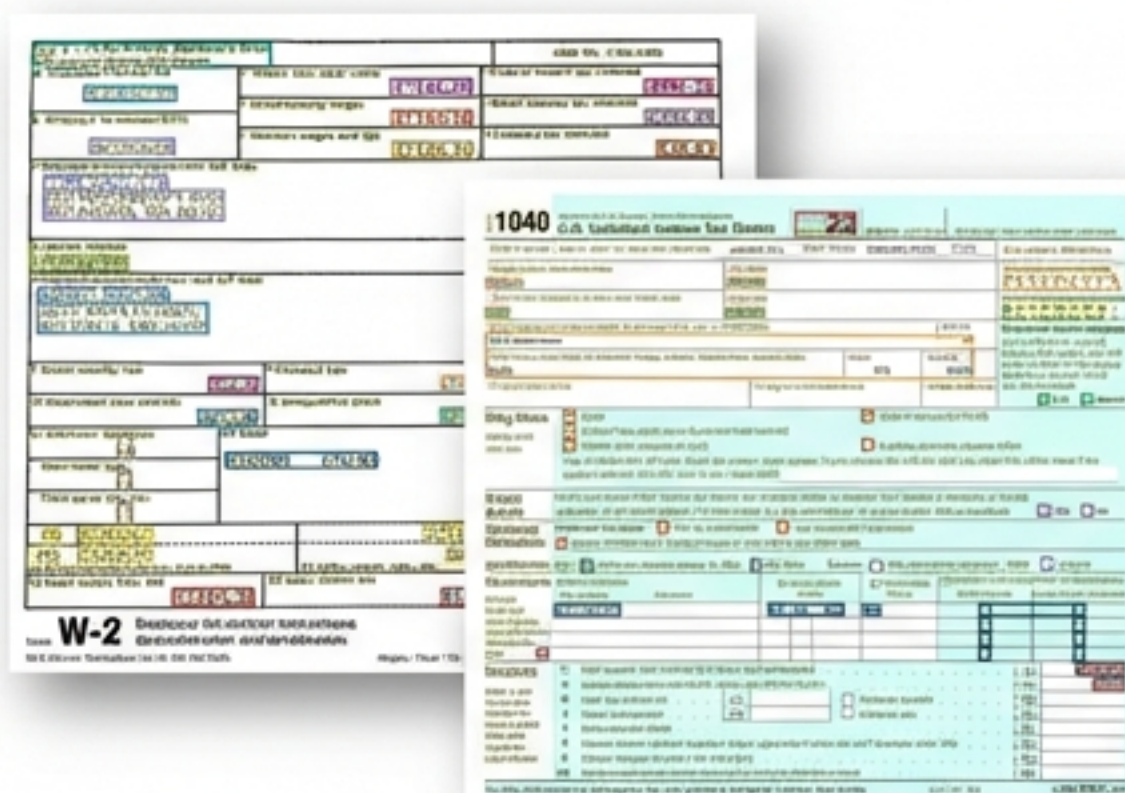
Compliance Ready: Tax, Identity & Legal Models

Identity Verification



Extracts Name, DOB, Expiration, Doc #. Covers US Licenses, Passports, Green Cards.

Unified US Tax Model



Single model handles W-2, 1098, 1099 variations, and 1040. Critical for Fintech/Proptech.

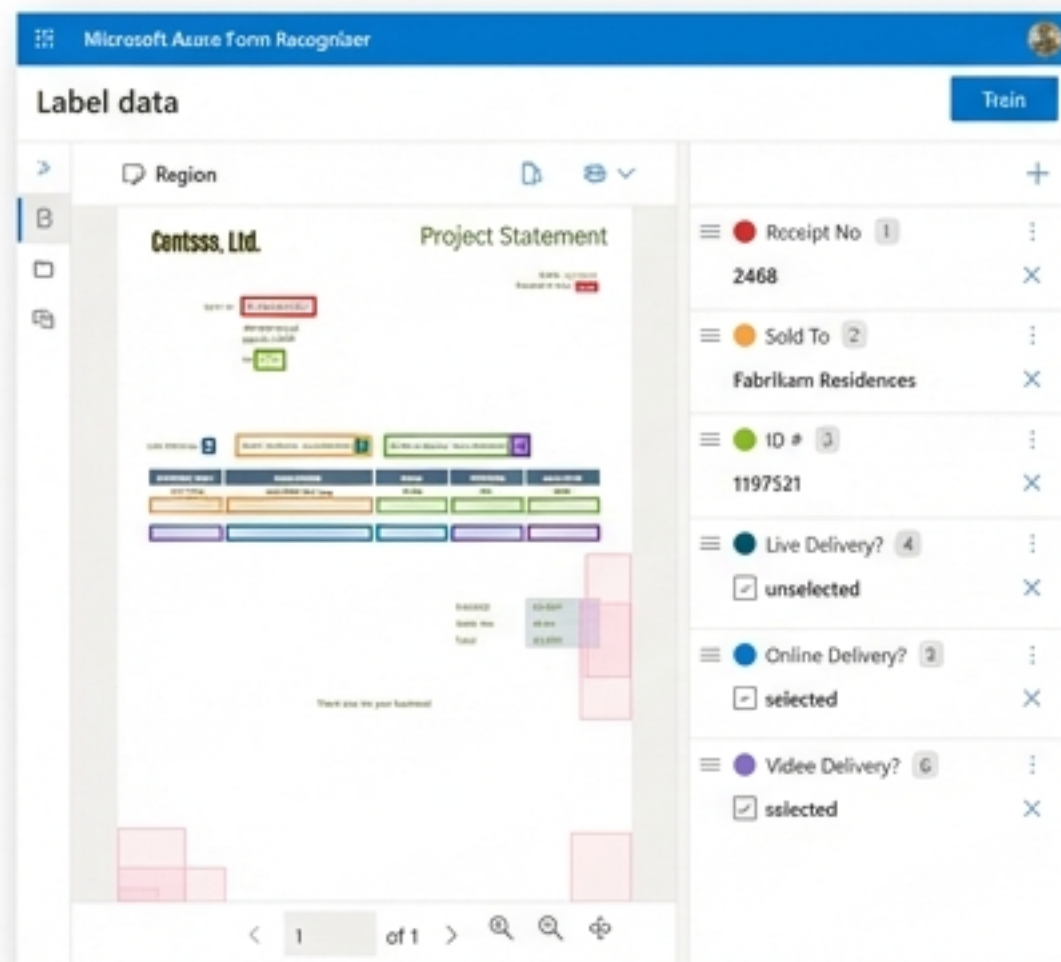
Legal & Admin

- Health Insurance Cards (Member ID, Group #)
- Contracts (Parties, Dates)
- Marriage Certificates

Custom Extraction: Beyond the Prebuilt

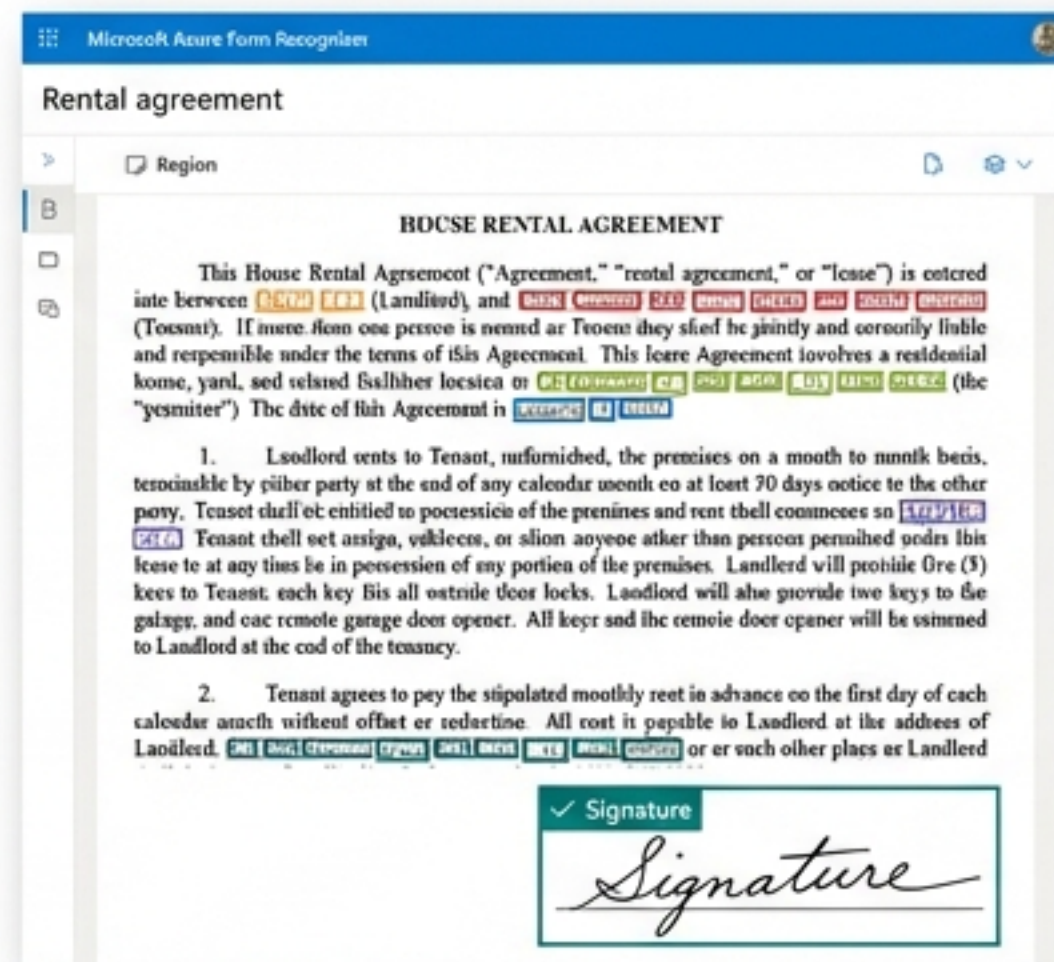
Custom Template

For static, consistent layouts (forms). Fast training, 50MB limit.



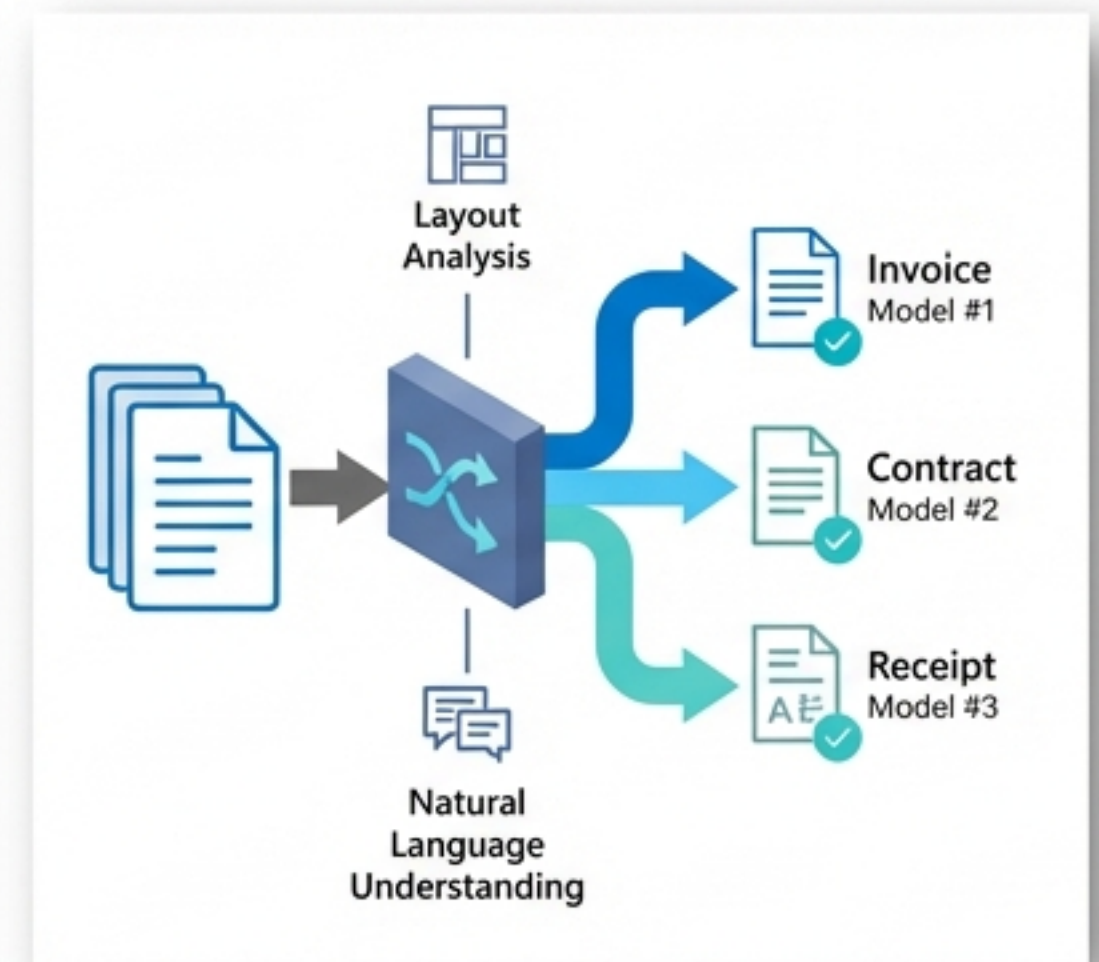
Custom Neural

For semi-structured/mixed docs.
Learns semantics.
New: Signature detection.



Custom Classification

Splits and identifies document types within a single file. Combines layout and language features.



- Composed Models allow grouping multiple models into one endpoint.

Global Reach & File Fidelity



Supported Inputs:

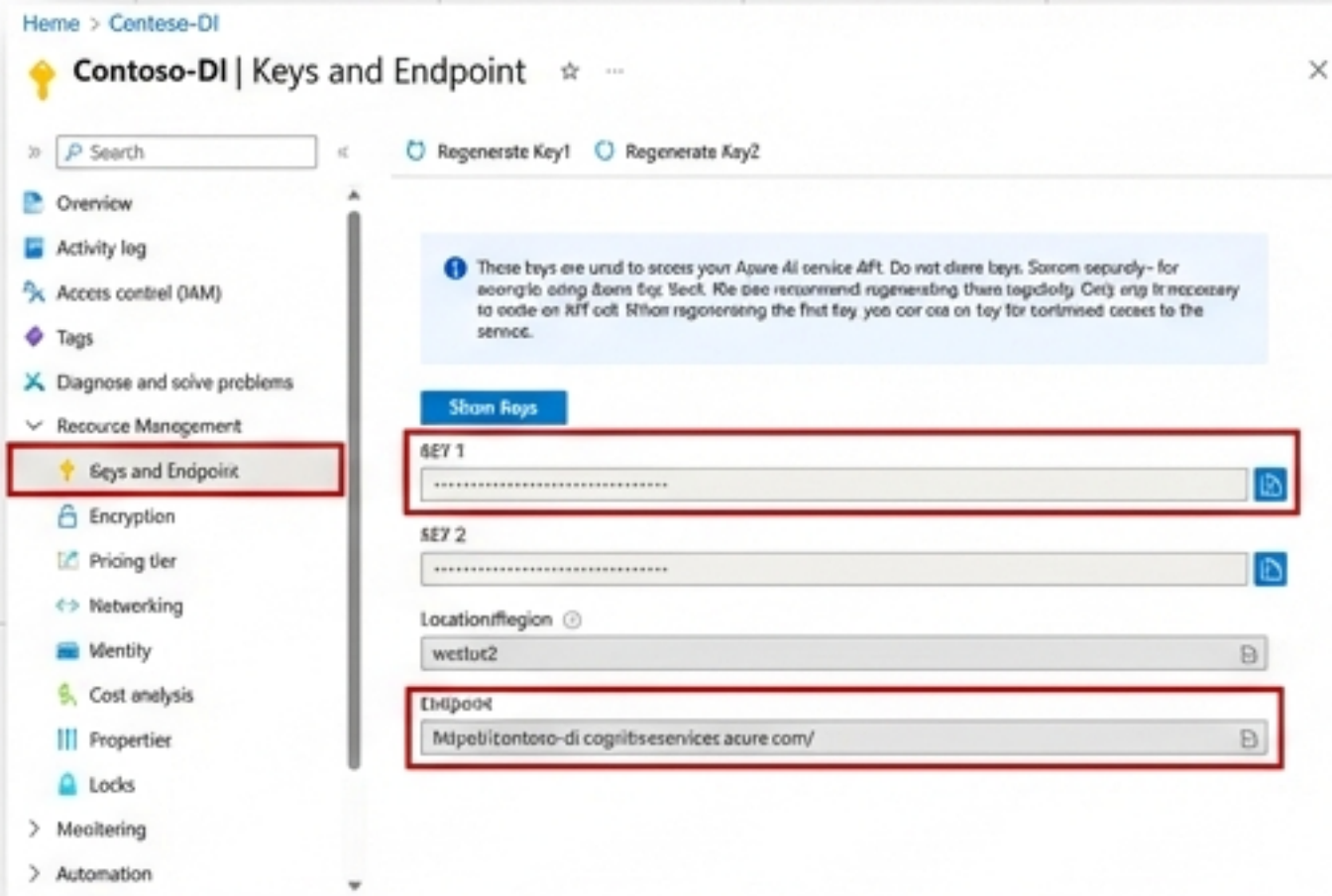
- Documents: PDF, HTML, Word (DOCX), Excel (XLS), PowerPoint (PPTX)
- Images: JPEG, PNG, BMP, TIFF, HEIF

Multilingual Capabilities:

- Print: Massive list including Cyrillic (Russian, Ukrainian), Arabic, Thai, Chinese, Japanese, Korean.
- Handwriting: Expanded to include Russian, Arabic, Thai.

Best Practice: Do not specify the language code unless necessary. The deep-learning technology auto-detects the text language for optimal results.

Developer Ecosystem & Integration



The Tooling

REST API v4.0 (2024-11-30 GA) | Document Intelligence Studio for visual testing.

The Code

```
var client = new DocumentAnalysisClient(new Uri(endpoint), new AzureKeyCredential(apiKey));
```

- C# / .NET
- Python
- Java
- JavaScript

Security, Privacy & Compliance



Data Privacy

Customer data policies ensure privacy. Models trained on your data remain yours and are not used to train base models.



Authentication

Managed Identities (Recommended).
Entra ID (RBAC).
API Keys (Secondary).



Network Security

Support for Virtual Networks (VNETs) and Private Endpoints.

Operational Limits & Capacity Planning

Service Tiers



Free (F0)

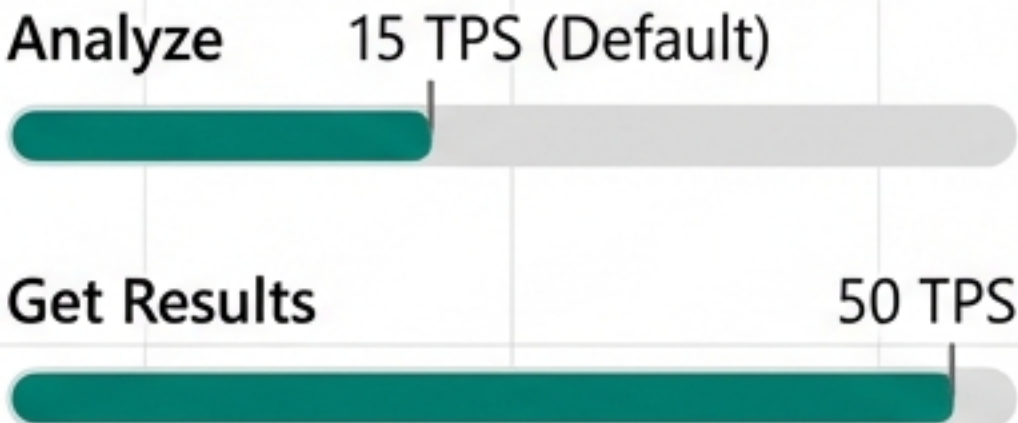
Progress bar
2 pages/min
500 pages/month



Standard (S0)

Adjustable limits
Adjustable limits
production scale

Throughput (TPS)



Throttling Strategy



Avoid sharp spikes. Ramp up gradually to allow autoscaling.

Model Limits

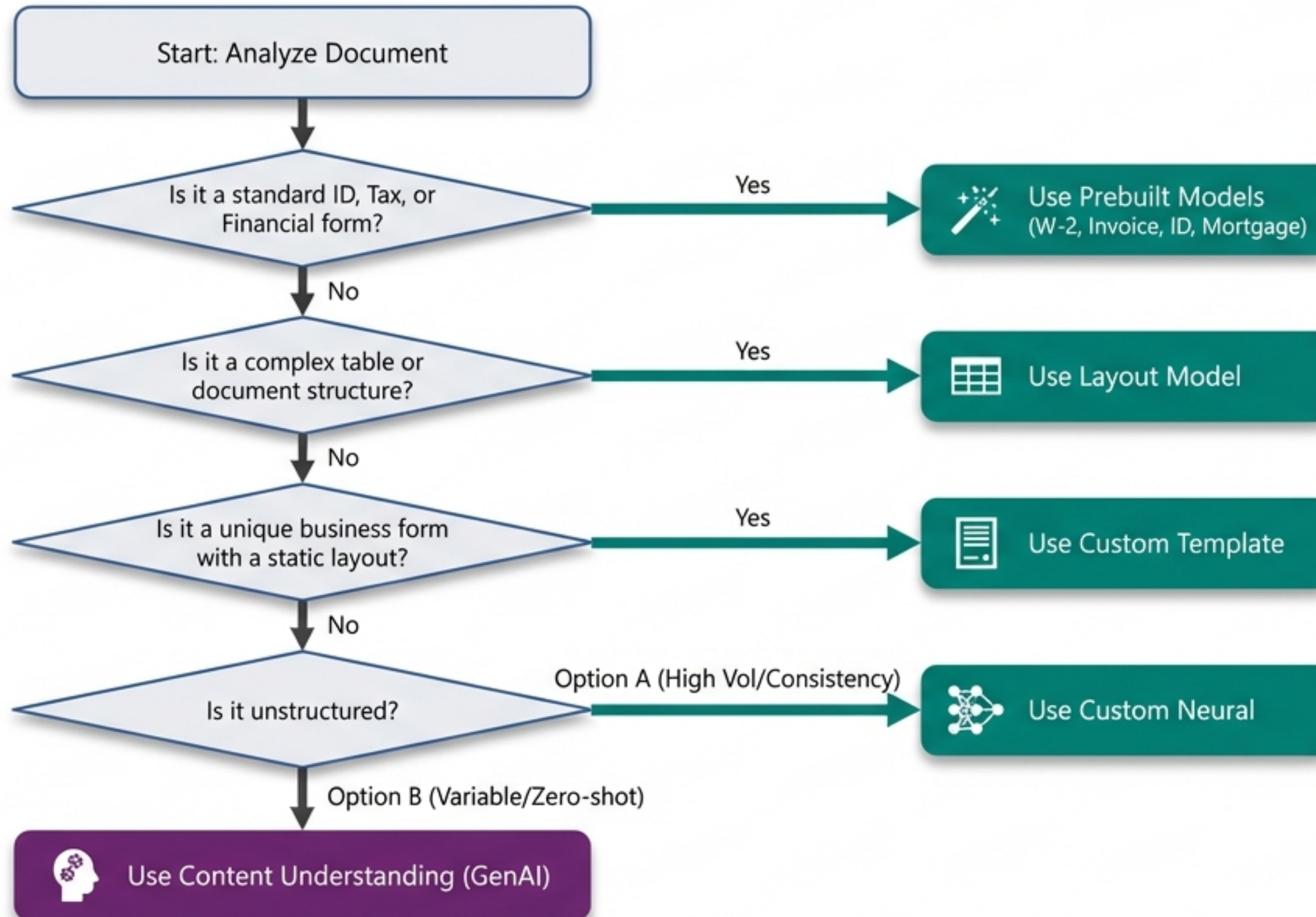


Custom Template:
Max 5,000 models



Custom Neural:
Max 500 models

The Model Decision Matrix



Resources & Next Steps

Documentation & Tools



Document Intelligence Studio:
formrecognizer.appliedai.azure.com



SDK References: Python, C#, Java, JS



Pricing Calculator: Pay-per-page vs.
Commitment tiers

Migration & Value



Migration Guide: Update from v3.1 to
to v4.0 (API version 2024-11-30).

**“Automate your data
processing with the power of
Traditional ML speed and
Generative AI reasoning.”**”

